Jacob Furtaw

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Baltimore, MD | Willing and Ready to Relocate Anywhere

Professional Summary

Results-driven ML Engineer & Agentic AI Builder with deep expertise in autonomous agents, multi-agent orchestration, and tool-calling systems. I specialize in transforming ambiguous requirements into production-ready agentic solutions through custom tool development, detailed system prompts, persistent memory/context management, and rigorous iteration — delivering reliable, observable systems that actually work in complex environments.

Skills

Agentic AI & Systems: Autonomous Agents • Multi-Agent Orchestration • Tool Calling & Reasoning Loops • Memory Management & Architectures • Advanced Prompt Engineering • RAG Pipelines • Observability & Transparency
Models & Inference: OpenAI, Anthropic, Grok, Gemini APIs • Local/Open-Weight Models (Ollama, vLLM, Llama.cpp, Nemotron Nano 3 Omni, Qwen2.5-Omni, Gemma) • Multimodal (Vision/Audio) • Quantization • Fine-Tuning
Frameworks & Tools: LangChain, Llama-Index, FastAPI, PyTorch, Transformers, Hugging Face, Docker, Git, Jupyter, NVIDIA NIMs & Foundation Models
Cloud & Deployment: Azure AI Foundry, Azure Resource Management, Azure ML Studio, Azure Container Apps

Work Experience

Agentic Engineer | Integral Federal | Hybrid — September 2026 - Present

- **Engineering an automated deep research agent** with web scraping, LLM relevance validation, structured data extraction, and automated export to Excel ledgers.
- **Architecting Azure cloud infrastructure**, managing VMs, container apps, pipelines, and Azure AI Foundry deployments to support multi-tenant agentic workflows.

Gen AI Engineer | Stealth AI Startup | Remote — January 2026 – September 2026

- Designing and building **production-grade local multi-agent systems** with advanced tool calling, memory management, and RAG for Family and Multi-Family Office clients, delivering reliable, privacy-first automation for complex financial and operational workflows.

Co-Founder & Machine Learning Engineer | SurgePoint Software | Remote — August 2023 – January 2025

- Led development of a production **RAG-powered AI student assistant** (Llama 3.1 / Mistral) serving 15k+ students with advanced prompt engineering and reliable agent performance.
- Built a **custom RAG pipeline** (Milvus) and curated a high-quality **13M-line dataset** from 200M+ lines of web-scraped data and thousands of documents — improved semantic relevance by **50-75%** and significantly reduced hallucinations.

Research Projects

Cloak AI - Privacy-First Local AI Agent

- Built a fully local multi-model AI agent with genuine tool use, mature RAG, dynamic model switching, and multimodal (text/image/audio) capabilities. Runs on consumer hardware.
- Implemented **inline real-time agent transparency/observability** for reasoning & tool-call monitoring
- Features include user-created knowledge bases, advanced web search tool, and inline image & stock data display.

Chat RAG & Supporting Projects| Project Link

- **Chat RAG:** Advanced modular RAG coding assistant supporting hundreds of local (Ollama/Hugging Face) and API-based models (OpenAI, Anthropic, NVIDIA). Includes dynamic model switching, custom prompts, quantization, and flexible data ingestion from local files, GitHub, and vector databases (**25 stars on GitHub**).
- **Agent Qwen**: Multimodal agent (Qwen2.5-Omni-7B) with voice I/O + image/video — Gradio, Whisper, PyTorch.

Automatic Identification of Equivalent Mutants using an ASTNN(GNN) | Project Link

- Delivered a transformer-based binary classification model (CodeBERT), improving F1 score and accuracy from 79% to 92% through hyperparameter tuning and dataset balancing techniques.
- Applied grid search and random search for hyperparameter optimization, along with over/undersampling methods to address class imbalance.

Education

Bachelor of Science in Computer Science, Software Engineering Concentration — December 2023
Towson University, Towson, MD